

**NATIONAL JUNIOR COLLEGE**

**Higher 2**



CANDIDATE  
NAME

BIOLOGY  
CLASS

REGISTRATION NO.

---

**BIOLOGY**

**9747/01**

JC2 Preliminary Examinations  
Paper 1

**11 Sept 2008**

Additional materials: Optical Answer Sheet

**1 Hour 15 minutes**

---

**READ THIS INSTRUCTIONS FIRST**

Write your name and registration number (e.g. 05IP0409 or 07S0620) and Biology Class (e.g. 2bi2E) clearly on the cover page.

Write your Name, Registration No. and Biology Class on your Optical Answer Sheet (OAS).

There are forty questions on this paper. Answer all questions.

For each question there are four possible answers A, B, C and D. Choose the one you consider correct and record your choice in 2B soft pencil on the Optical Answer Sheet.

Each correct answer will score one mark. A mark will NOT be deducted for a wrong answer.

Any rough working should be done in this booklet.

This paper consists of **18** printed pages including cover page.

- 1 Which of the following best accounts for the difference in structure between amylose which is helical and cellulose which is fibrous?
  - A Presence or absence of  $180^\circ$  rotation of alternate subunits.
  - B Presence or absence of branching in the macromolecule
  - C Subunits are either monosaccharides or amino acids.
  - D Different tendency of macromolecule to form hydrogen bonds with water.
  
- 2 The molecule shown below is found in various parts of the human body, and is helical in structure.

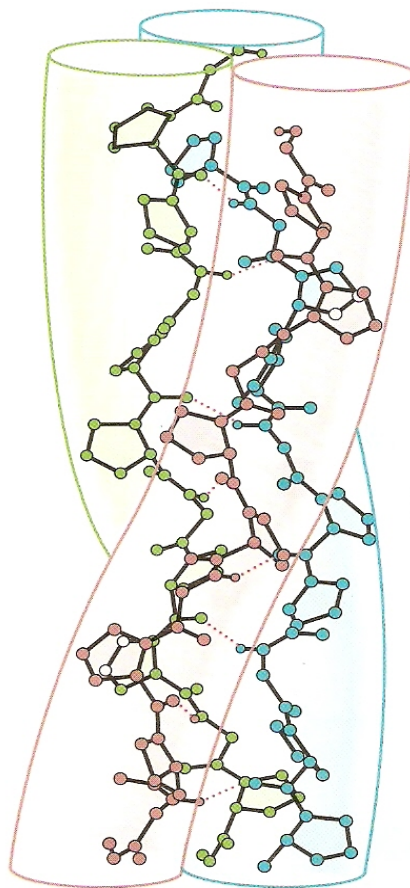
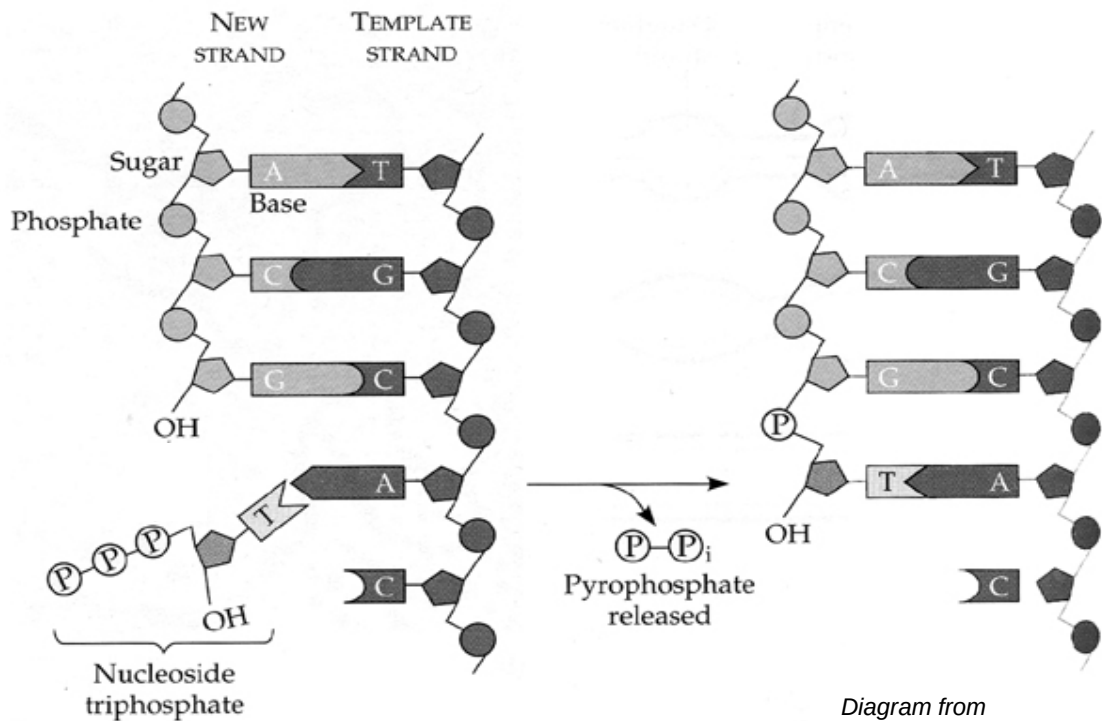


Diagram from:  
 "Biochemistry" by Garrett and Grisham

What else is true of this molecule?

- A Complementary base pairing occurs through the formation of hydrogen bonds between chains.
- B Hydrogen bonds within each chain give each chain its helical structure.
- C Glutamine occurs very frequently in the sequence of the subunits in the molecule.
- D The entire molecule interacts with proteins to form complexes necessary for the control of cell activities.

- 3 The diagram below shows the synthesis of a new strand of DNA during interphase.

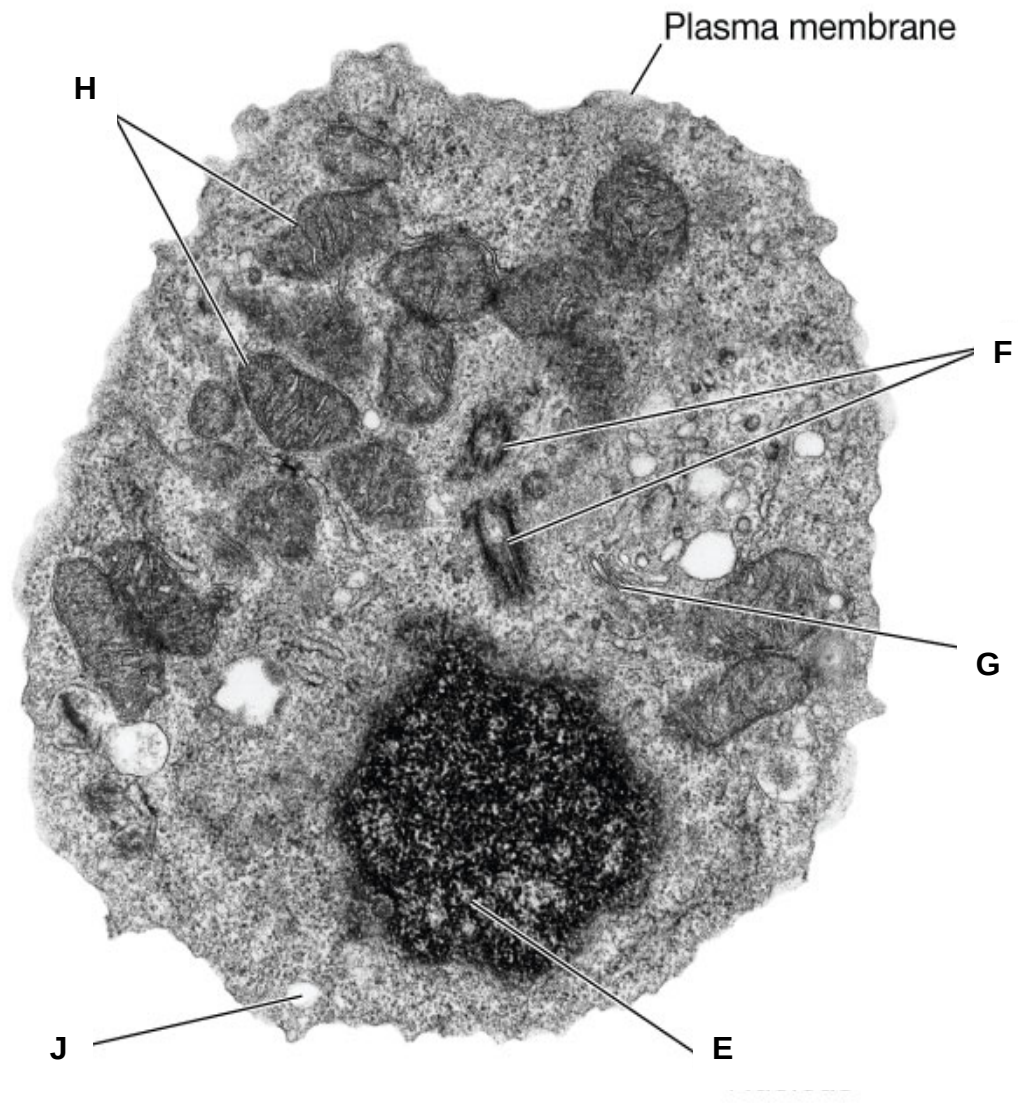


Which of the following shows the correct combination of bond(s) that need to be formed and the kind of reaction that is involved in order for the nucleotide to be added to the DNA chain?

|   | Bond(s) to be formed      | Reaction(s) involved               |
|---|---------------------------|------------------------------------|
| A | Phosphodiester            | Condensation and Dephosphorylation |
| B | Phosphodiester            | Hydrolysis                         |
| C | Phosphoester and Hydrogen | Condensation and Hydrolysis        |
| D | Phosphoester              | Dephosphorylation                  |

- 4 Which of the following is / are true of triglycerides and phospholipids?
- 1 Both contain glycerol
  - 2 Both are amphipathic
  - 3 Triglycerides have a higher carbon content than phospholipids for storage purposes
  - 4 Phospholipids have hydrophilic regions to interact with cell cytoplasm unlike triglycerides
  - 5 Triglycerides are formed from glycerol and fatty acids but phospholipids are formed from glycerol and phosphoric acid.
- A 1, 3 and 5
- B 1, 2 and 3
- C 1, 3 and 4
- D 2, 3 and 4
- 5 A mitochondrion in a liver cell measured  $1.2\text{ }\mu\text{m}$ . On an electron micrograph the length of this organelle was 36mm. What is the magnification of the electron micrograph?
- A 3000 times
- B 12000 times
- C 15000 times
- D 30000 times
- 6 Which of the following statements about the cell is true?
- A Endocytosis is an active process because energy is required to move materials into the cell against their concentration gradients.
- B In simple diffusion, materials enter the cell down their concentration gradients whereas facilitated diffusion can move materials against their concentration gradients.
- C In osmosis, water, ions and small polar molecules enter the cell by diffusing through a selectively permeable membrane down their concentration gradients.
- D Materials such as ions and small polar molecules can enter the cell against their concentration gradient by active transport.

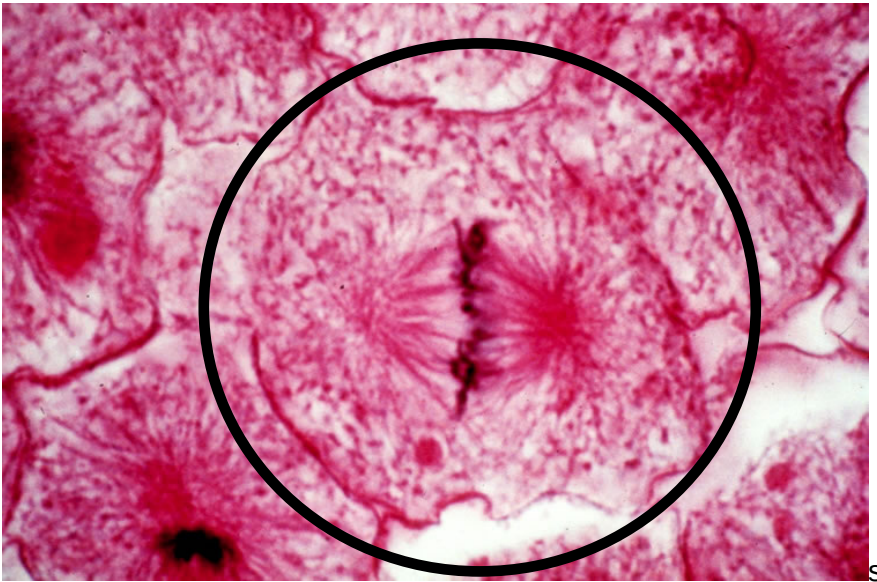
7 An electron micrograph of a cell is shown below.



Match the organelles **E, F, G, H** and **J** associated with the cellular processes listed.

|          | <b>E</b>                  | <b>F</b>              | <b>G</b>                        | <b>H</b>                        | <b>J</b>                        |
|----------|---------------------------|-----------------------|---------------------------------|---------------------------------|---------------------------------|
| <b>A</b> | DNA replication           | Digestion of material | Organizes the spindle           | Oxidative phosphorylation       | Packaging of secretory products |
| <b>B</b> | Oxidative phosphorylation | Organizes the spindle | Digestion of material           | DNA replication                 | Packaging of secretory products |
| <b>C</b> | Organizes the spindle     | Digestion of material | Oxidative phosphorylation       | Packaging of secretory products | DNA replication                 |
| <b>D</b> | DNA replication           | Organizes the spindle | Packaging of secretory products | Oxidative phosphorylation       | Digestion of material           |

- 8 Which of the following statement about enzymes is inaccurate?
- A Enzyme function is not influenced by factors such as lead nitrate.
  - B Enzyme function is reduced if the three-dimensional structure or conformation of an enzyme is altered.
  - C Enzymes increase the rate of chemical reaction by lowering activation energy barriers.
  - D Enzymes may require a non-protein cofactor or ion for catalysis to take place.
- 9 The circled image below shows a stage of cell division in the white fish blastula.



Choose the most appropriate description that depicts what was happening.

- A Attachment of spindle fibre, chromosomes fully condensed, separation of sister chromatids
- B Chromosomes line up, attachment to centrioles, furrowing of cell membrane
- C Chromosomes positioned in the centre, to be pulled towards opposite poles, furrowing of cell membrane
- D Chromosomes fully condensed, line up in the plate of the cell, asters formed

**10** Which of the following is unique to mitosis and not a part of meiosis?

- A** Homologous chromosomes pair forming bivalent.
- B** Homologous chromosomes cross over.
- C** Chromatids are separated during anaphase.
- D** Homologous chromosomes behave independently.

**11** A couple has a son who has blood group A and phenylketonuria. One parent has blood group O and the other has blood group AB. Neither parent has phenylketonuria.

What is the probability that the second child of the couple will be a girl with blood group B and does not have phenylketonuria ?

- A** 1 in 16                      **B** 3 in 16                      **C** 1 in 8                      **D** 3 in 8

**12** In fruit flies, body colour and eye colour are governed by different gene loci. A cross between yellow-bodied white-eyed females and grey-bodied red-eyed males produced females which were grey-bodied and red-eyed. When the  $F_1$  females were backcrossed with their male parents, the resulting  $F_2$  males comprised the following:

1330 yellow-bodied white-eyed flies,  
26 yellow-bodied red-eyed flies,  
35 grey-bodied white-eyed flies, and  
1297 grey-bodied red-eyed flies

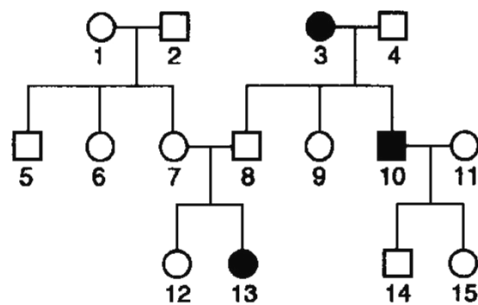
Which of the following explains the above results?

- A** In fruit flies, the male is the homogametic sex.
- B** The two gene loci segregate by independent assortment.
- C** The two gene loci are sex-linked and crossing-over had taken place during gamete formation.
- D** The allele for grey body is dominant over the allele for yellow body and the allele for white eyes is dominant over the allele for red eyes.

- 13 The genotypes of a tomato variety were **TtHh**, **Tthh**, **tHh** and **tthh**, where **T** = tall and **H** = hairy. Which one of the following crosses would produce progeny giving a phenotypic ratio approximating to 1:1:1:1?

- A **TtHh** x **TtHh**
- B **TtHh** x **Tthh**
- C **TtHh** x **tthh**
- D **TTHH** x **tthh**

- 14 The family tree shows the inheritance of a condition caused by the recessive allele **r**.



key

- |                 |                   |
|-----------------|-------------------|
| ○ normal female | ● affected female |
| □ normal male   | ■ affected male   |

Which of the females are certain to have the genotype **Rr**?

- A 7, 9 and 15
  - B 1, 7 and 12
  - C 1, 6 and 7
  - D 9, 12 and 15
- 15 What type of natural selection is occurring when the average phenotype is selected for and the extreme phenotypes are selected against?

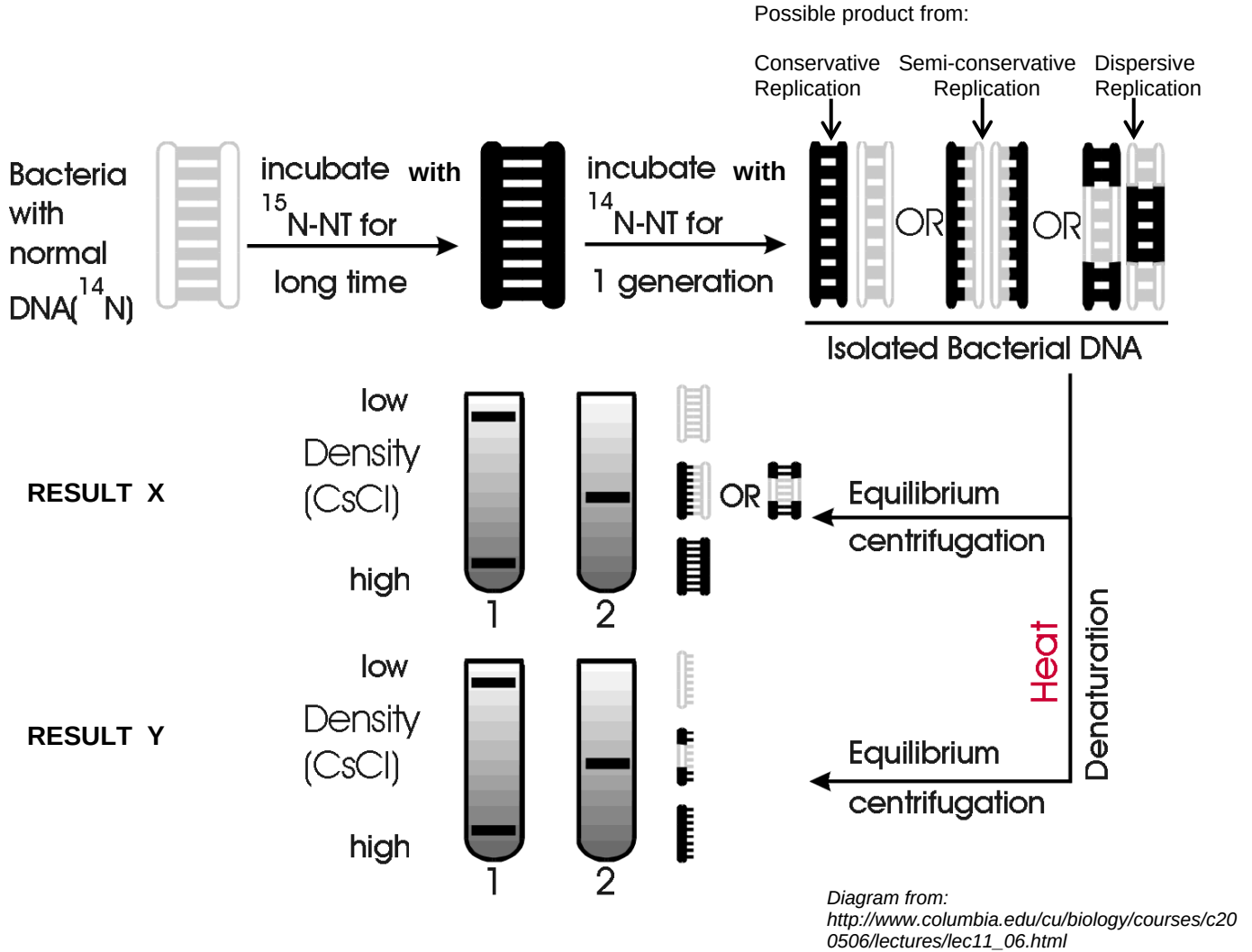
- A disruptive
- B directional
- C stabilising
- D reductive



- 16** Which of the following best describes the type of fossils that are found in the deepest strata of the earth?
- A** Old and complex
  - B** Old and simple
  - C** Young and complex
  - D** Young and simple
- 17** Which one of the following statements is related to the principle of adaptive radiation in evolution?
- A** Unrelated organisms living in similar environments may tend to resemble each other.
  - B** Related organisms living in different environments show modifications of an underlying unity of pattern.
  - C** Selection in a constant environment tends to maintain the most frequently occurring variety of a species.
  - D** Species evolve by selection altering the gene frequencies in interbreeding populations.
- 18** What do chromosomal aberrations and gene mutations have in common and how are they different?

|          | Similarity                                                 | Difference                                                                                         |
|----------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>A</b> | Both may involve addition of nucleotides.                  | Gene mutations always produce dominant alleles but not chromosomal aberrations.                    |
| <b>B</b> | Both may not result in disorders.                          | Gene mutations do not involve inversions but inversion of segments of chromosomes do occur.        |
| <b>C</b> | Both affect DNA sequence.                                  | Gene mutations occur within a chromosome but chromosomal aberrations may occur across chromosomes. |
| <b>D</b> | Both may not result in a difference in protein expression. | Chromosomal aberrations are more harmful than gene mutations.                                      |

- 19 The Meselson-Stahl experiment is shown below. This experiment was done to prove that replication is via a semi-conservative mode rather than a conservative or dispersive mode.



Which result (tube 1 or 2) should be obtained, at each centrifugation X and Y, given that replication is semi-conservative? Which type of replication does the result obtained from centrifugation X and Y effectively eliminate?

|          | Centrifugation X |              | Centrifugation Y |              |
|----------|------------------|--------------|------------------|--------------|
|          | Correct result   | Eliminates   | Correct result   | Eliminates   |
| <b>A</b> | 1                | Dispersive   | 1                | Conservative |
| <b>B</b> | 2                | Conservative | 2                | Dispersive   |
| <b>C</b> | 1                | Dispersive   | 2                | Conservative |
| <b>D</b> | 2                | Conservative | 1                | Dispersive   |

Refer to the diagram below to answer questions 20 and 21.

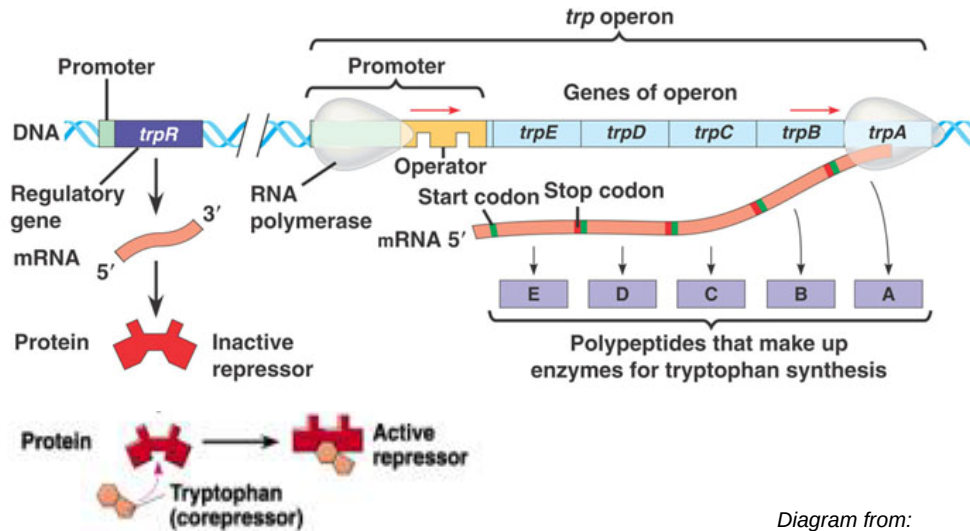


Diagram from:  
"Biology" by Campbell & Reece

20 If the *trp* operon were to be **unable to produce any enzymes** regardless of the presence or absence of the amino acid tryptophan, what could be the likely reason for this?

- A Tryptophan is always bound to repressor.
- B Operator sequence has been deleted.
- C Repressor is unable to bind to operator.
- D *trpR* has been deleted.

21 How is the *trp* operon different from the *lac* operon?

*trp* operon

*lac* operon

- |                                                      |                                                |
|------------------------------------------------------|------------------------------------------------|
| A Produces polycistronic mRNA                        | Produces monocistronic mRNA                    |
| B Repressor binds to operator                        | Activator binds to operator                    |
| C Repressible                                        | Inducible                                      |
| D Transcription occurs in the presence of tryptophan | Transcription occurs in the absence of glucose |

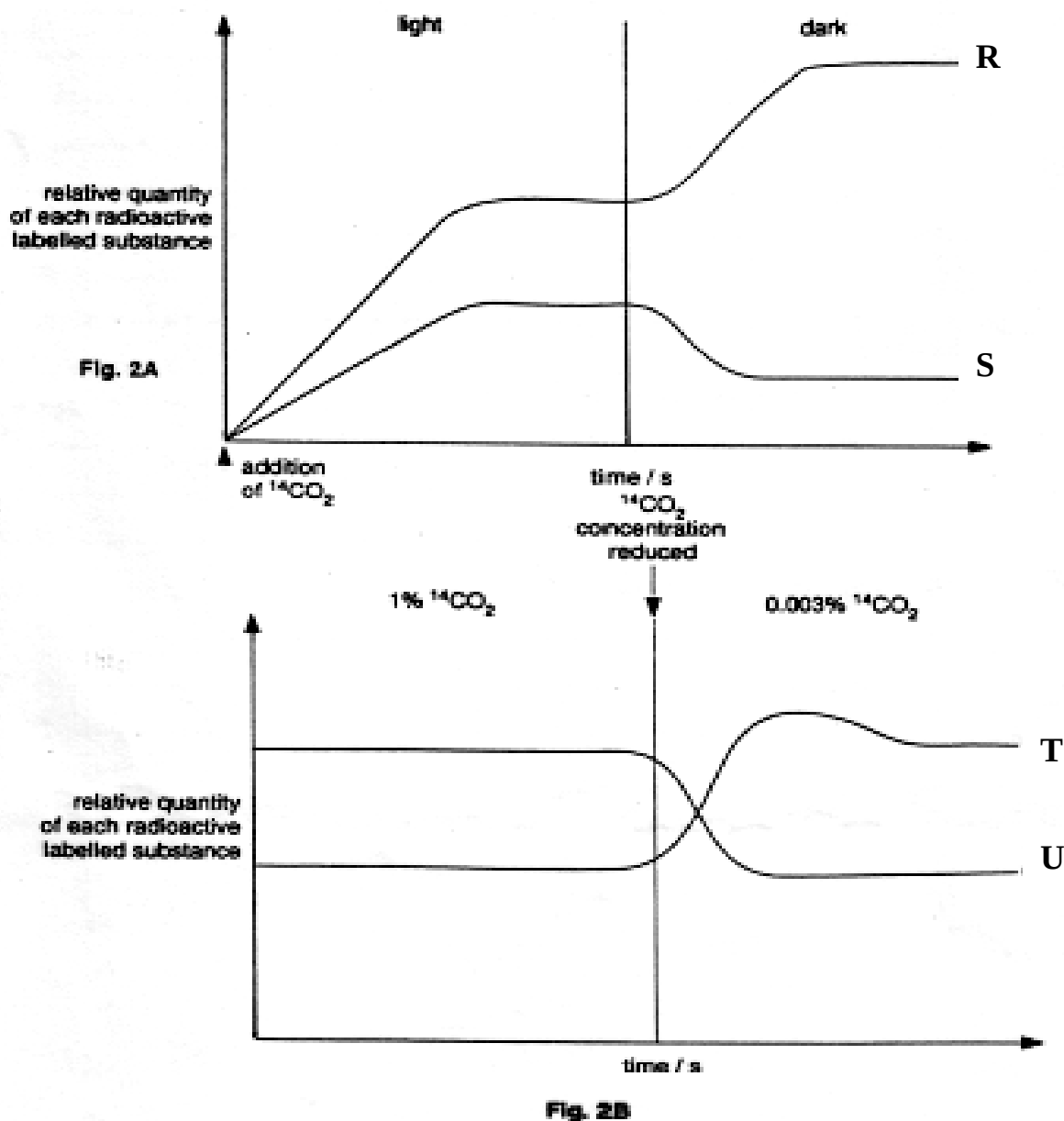
22 Which of the functions of RNA is incorrectly matched?

- A Micro RNAs bind with mRNA to inhibit gene expression.
- B Ribosomal RNAs bind with tRNA to catalyse the formation of phosphodiester bonds.
- C Small nuclear RNAs bind with ribonucleoproteins to form spliceosomes.
- D Transfer RNAs bind with mRNA to facilitate translation.

- 23** Repetitive sequences do not have the function of
- A** binding to specific transcription factors
  - B** being transcribed into RNA
  - C** coding for functional proteins
  - D** being used for DNA fingerprinting
- 24** The basis for development of leukemia in some children treated with a retrovirus-derived gene therapy vector is:
- A** Insertion of the viral genome adjacent to an oncogene
  - B** Use of a virus that contains an oncogene
  - C** Insertion causing a mutation of a tumor suppressor gene
  - D** Stimulation of immune response to cell surface antigens introduced by an inserted gene
- 25** Which of the following statements about proto-oncogenes is true?
- A** Increase in amount of RAS protein production may be a result of gene amplification
  - B** Loss of function mutations tend to occur in proto-oncogenes
  - C** Switching of the position of proto-oncogene will not increase the activity of the transcription
  - D** They are originally derived from RNA tumour virus
- Will be A on final
- 26** Which of the following would be the first step in the biosynthesis of a virus with reverse transcriptase?
- A** A complementary strand of DNA must be synthesized from an DNA template
  - B** Double-stranded RNA must be synthesized
  - C** A complementary strand of DNA must be synthesized from an RNA template
  - D** A complementary strand of RNA must be synthesized.

- 27 Which of the following statements about bacteria is true?
- A Bacteria possess 80s ribosomes , introns and RNA processing
  - B Bacteria possess 70s ribosomes , introns and no RNA processing
  - C Bacteria possess 70s ribosomes , no introns and no RNA processing
  - D Bacteria possess 80s ribosomes , RNA processing and no introns
- 28 What is present in a bacteriophage but not in an influenza virus?
- A DNA
  - B RNA
  - C Glycoprotein
  - D Viral capsid
- 29 In which of the following changes is most energy released during the metabolism of carbohydrates?
- A Maltose to glucose
  - B Glucose to phosphoglyceraldehyde
  - C Phosphoglyceraldehyde to pyruvic acid
  - D Pyruvic acid to carbon dioxide and water
- 30 Which one of the following is a correct outline of the main events in photosynthesis ?
- A Light joins carbon dioxide to an acceptor compound which is then reduced by hydrogen obtained from water.
  - B Carbon dioxide combines with an acceptor compound and this is reduced by hydrogen split from water by light
  - C Light splits carbon dioxide and the resulting carbon then combines with oxygen and hydrogen obtained from water.
  - D Light splits water and the resulting hydroxyl group combines with a compound which has incorporated carbon dioxide

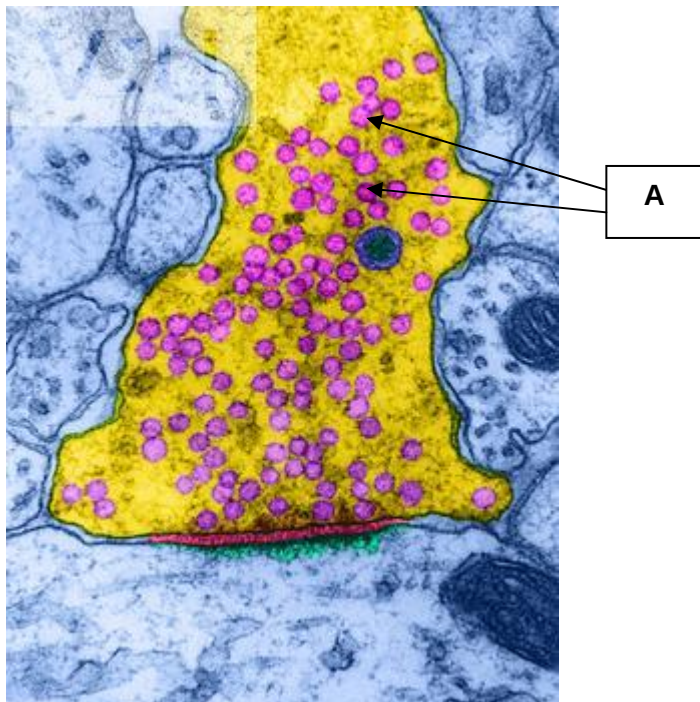
- 31 Fig. 2A and Fig. 2B show the results of two separate experiments on carbon dioxide fixation in photosynthesis using a unicellular green alga and the radioactive isotope  $^{14}\text{C}$  to label carbon dioxide. The algae were actively photosynthesizing before the start of both experiments.



With reference to figures 2A and 2B, identify the graphs **R**, **S**, **T**, and **U**.

|          | <b>R</b> | <b>S</b> | <b>T</b> | <b>U</b> |
|----------|----------|----------|----------|----------|
| <b>A</b> | RuBP     | GP/PGA   | GP/PGA   | RuBP     |
| <b>B</b> | GP/PGA   | RuBP     | GP/PGA   | RuBP     |
| <b>C</b> | GP/PGA   | RuBP     | RuBP     | GP/PGA   |
| <b>D</b> | RuBP     | GP/PGA   | RuBP     | GP/PGA   |

- 32 The diagram below shows part of an electron micrograph of a neuron. Name the portion of the neuron that forms the membrane which fuses with structures **A**.



- A** Terminal button  
**B** Axon  
**C** Dendrite  
**D** Synapse
- 33 Studies on the giant axon of the squid showed that bacteria toxin BITIX results in an increase in negativity of the postsynaptic membrane. What is the possible mode of action of BITIX?
- A** The release of acetylcholine  
**B** Analog of acetylcholine  
**C** Release of potassium ions from postsynaptic membrane  
**D** Mimicking of cholinesterase

**34** What is not necessary for PCR to occur?

- A** dATP
- B** Primers
- C** DNA fragments
- D** Ribonucleotides

**35** Anti-thrombin (a protein involved in blood clotting) and human growth hormone are examples of human proteins that can be produced via the genetic engineering technique.

How would the process which produces anti-thrombin differ from the one that produces a bacterial antibiotic resistance protein if both were to be carried out in bacteria?

- A** Antibiotic resistance would not be used as a selection marker for successfully transformed bacteria for the process that produces the antibiotic resistance protein.
- B** The Anti-thrombin gene insert would require an additional eukaryotic promoter.
- C** Anti-thrombin would require additional post-translational modification.
- D** Different plasmids would have to be used.

**36** Which of the following is are reasons for scientists to employ the method of plant cloning?

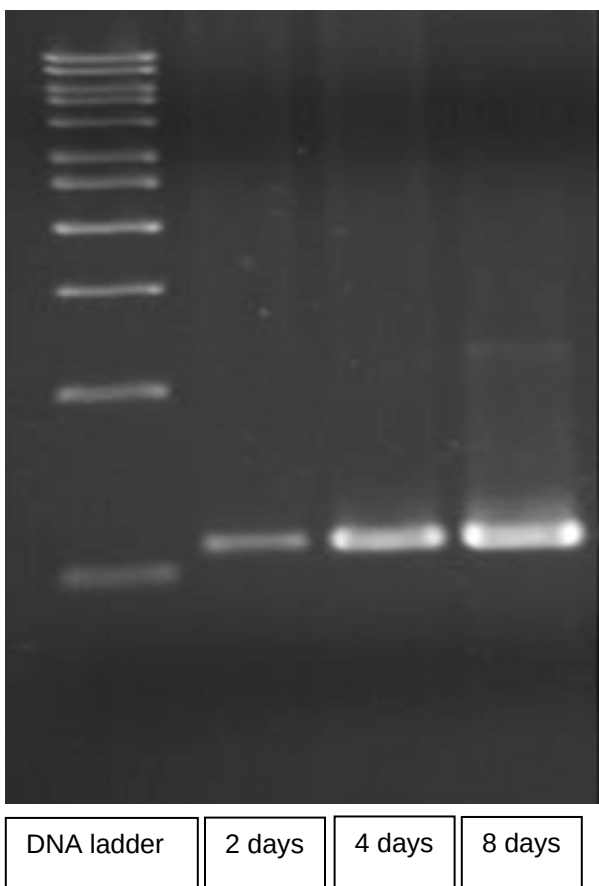
- 1** change the phenotype of an organism
- 2** create smaller genetic changes at a much more rapid pace
- 3** production of large amount of pharmaceuticals
- 4** selective breeding is too slow

- A** 1 and 2
- B** 1 and 3
- C** 1, 2 and 3
- D** 1, 2, 3 and 4



- 37** The sample shows the amount of purified gene M isolated from single cells obtained from an embryo after 2 days, 4 days and 8 days of development.

Which type of gene regulation is this likely to illustrate?



- A** Gene amplification
- B** DNA methylation
- C** Histone acetylation
- D** Upregulation of transcription
- 38** The genetically engineered super salmon was created from Atlantic salmon stocks and are capable of growing to a large size in 14 months. Which of the following is not a benefit intended from the crop?
- A** Higher yield for farmers
- B** Minimising pollution
- C** Decreasing the food consumption of the crops in their lifetime
- D** Increase in supply to meet world's demand

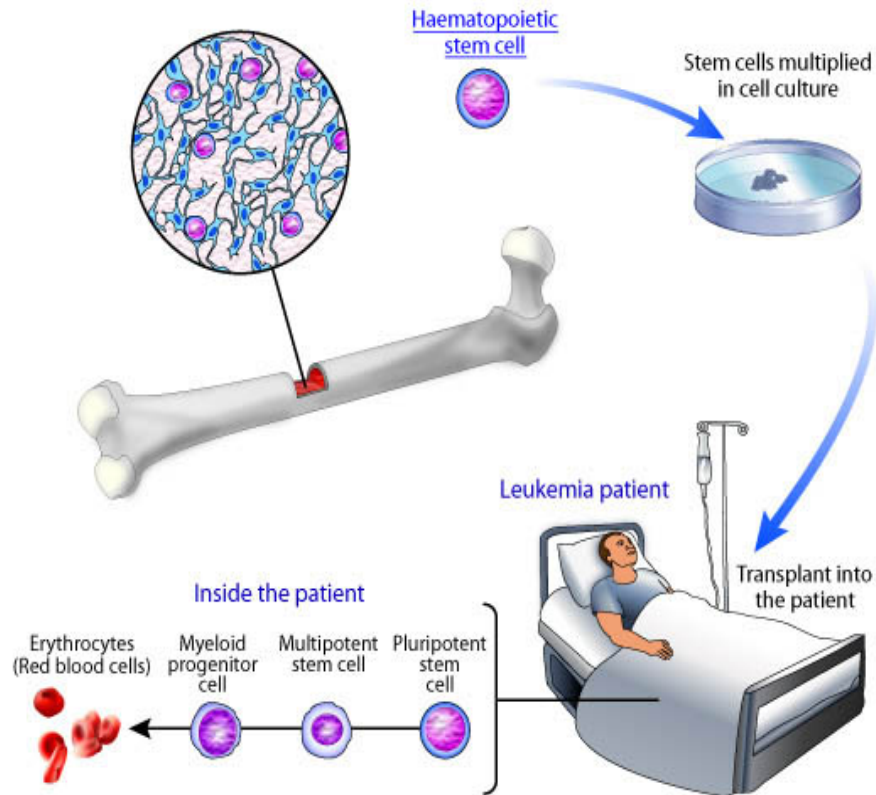


Illustration by [Cell Imaging Core](#) of the Center for Reproductive Sciences.

Which of the following options is not a characteristic of the form of treatment shown above?

- 1 Easy to extract in laboratory
- 2 Readily available
- 3 Difficult to control
- 4 Limited longevity

A 1 only                      B 1 and 2                      C 1 and 3                      D 1, 2, 3 and 4

40 Germ-line gene therapy has not been prevalent in medical treatment because of many reasons. Which of the following is/are implications of germ-line therapy?

- 1 unforeseeable clinical risks
- 2 opens problems for social discrimination
- 3 cost
- 4 issue of consent

A 1 only                      B 2, 3 and 4                      C 1, 2 and 4                      D 1, 2, 3 and 4

--- End of Paper ---

## Answers to H2 Prelim Paper 1

|    |   |    |   |    |   |    |   |
|----|---|----|---|----|---|----|---|
| 1  | A | 11 | B | 21 | C | 31 | C |
| 2  | B | 12 | C | 22 | B | 32 | A |
| 3  | C | 13 | C | 23 | A | 33 | C |
| 4  | C | 14 | A | 24 | C | 34 | D |
| 5  | D | 15 | C | 25 | A | 35 | C |
| 6  | D | 16 | B | 26 | C | 36 | D |
| 7  | D | 17 | B | 27 | C | 37 | A |
| 8  | A | 18 | C | 28 | A | 38 | B |
| 9  | D | 19 | D | 29 | D | 39 | B |
| 10 | D | 20 | A | 30 | B | 40 | D |